Classification Metrics

Tim Book

GENERAL ASSEMBLY

Agenda

- [ ] Differences between regression and classification metrics
- [ ] Single-number metrics and the confusion matrix
- [ ] The ROC curve and the AUC score
- [ ] Python examples!

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Classification vs Regression Metrics

Why do we need a whole new set of metrics?

For **regression**, we're primarily concerned with how close our predictions are to the truth. This was generally some clever summarization of the **residuals**. That is, some measurement of $y - \hat{y}$.

But for **classification**, we don't have residuals! There's no numeric difference between category variables.

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Classification Metrics: Framing

Up until now, we've only used **accuracy** as our evaluation metric. But it many cases, especially those with **class imbalance**, it can be a poor (or misleading!) metric.

For the first time, we need to consider the actual context of the problem when deciding on the most important evaluation metric.

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Some Quick Vocab

Suppose you build a logistic regression model to predict whether or not someone will vote in the upcoming election. You make predictions for 100 people and then follow up with them:

- There are 40 people you predicted to vote who actually voted.
  -
- There are 20 people you predicted to vote but did not vote.
  -
- There are 15 people you predicted to stay home who actually voted.
  -
- There are 25 people you predicted to stay home who did not vote.
  -

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  - These are **true negatives**

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Some Quick Vocab

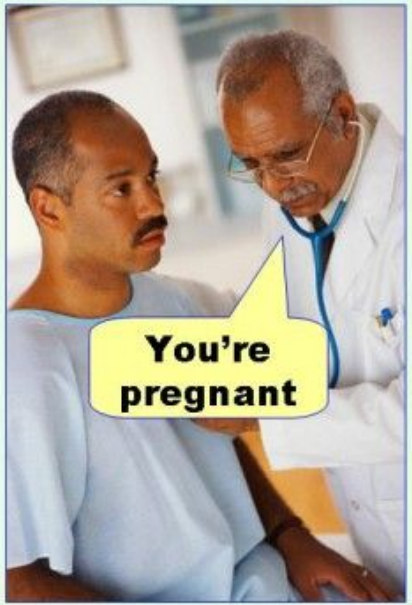
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- There are 40 people you predicted to vote who actually voted.
  - These are **true positives**
- There are 20 people you predicted to vote but did not vote.
  - These are **false positives**
- There are 15 people you predicted to stay home who actually voted.
  - These are **false negatives**
- There are 25 people you predicted to stay home who did not vote.
  - These are **true negatives**

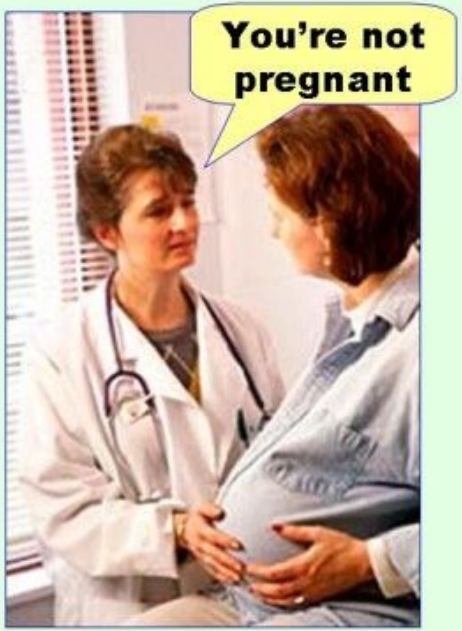
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Another Name for These

Type I error
(false positive)



Type II error
(false negative)



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A Simple Mnemonic





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Aside on Type I and II Errors

In the fields of inference and decision theory, Type I and Type II errors are used more commonly. We can actually match some terms up here:

	H_0 True	H_0 False
Fail to Reject H_0		
Rejected H_0		

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	H_0 True	H_0 False
Fail to Reject H_0	*(No catchy name)*	β
Rejected H_0	α	Power (K)

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Example: Cancer Detection

You are a physician doing cancer screens. You conduct a screen and must either report “cancer” (a positive) or “no cancer” (a negative).



What would be considered a false positive? A false negative? Which is worse?

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Example: Advertising

You run a digital ad agency and you're deploying a model that determines whether or not you should show an ad to the visitor of a website based on metadata you've collected about this user. You either show the specific ad for your client (positive) or you show a random other ad from your portfolio (negative). Your goal is that the user will click this ad.



What would be considered a false positive? A false negative? Which is worse?

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The Confusion Matrix and Accompanying Statistics

The Confusion Matrix

A **confusion matrix** is a two-way frequency table showing the counts of all of your outcomes in terms of predicted vs actual:

	Actual Positive	Actual Negative
Predicted Positive		
Predicted Negative		

(Note that some software, including Scikit-Learn, will flip the rows and columns. Pay attention!)

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The Confusion Matrix

A **confusion matrix** is a two-way frequency table showing the counts of all of your outcomes in terms of predicted vs actual:

	Actually voted	Did not vote
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(Note that some software, including Scikit-Learn, will flip the rows and columns. Pay attention!)

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Classification Metrics

The confusion matrix can gives us a bird's-eye view of performance, but we can also use it to calculate some single-number summaries:

- Accuracy
- Misclassification rate
- Sensitivity
- Specificity
- Precision
- Recall

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Accuracy

The **accuracy** of a model is the proportion of correct predictions it makes.

The **misclassification rate** is the opposite - the proportion it gets incorrect!

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Accuracy

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	Actually voted	Did not vote
Predicted to vote	40	20
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What is the accuracy of our voter model?

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Accuracy

	Actually voted	Did not vote
Predicted to vote	40	20
Predicted stay home	15	25

$$\text{Acc} = \frac{40+25}{40+20+15+25} = \frac{65}{100} = 0.65$$

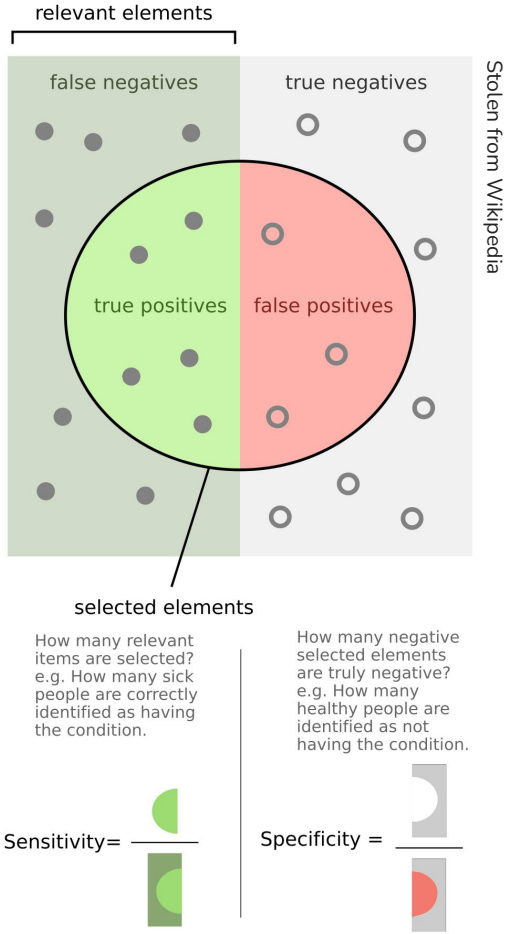
$$\text{Miscl. Rate} = \frac{20+15}{40+20+15+25} = \frac{35}{100} = 1 - 0.65 = 0.35$$

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Sensitivity & Specificity

The **sensitivity** is the **true positive rate** - that is, among all true positives, what proportion did we detect?

The **specificity** is the **true negative rate** - that is, among all true negatives, what proportion did we detect?



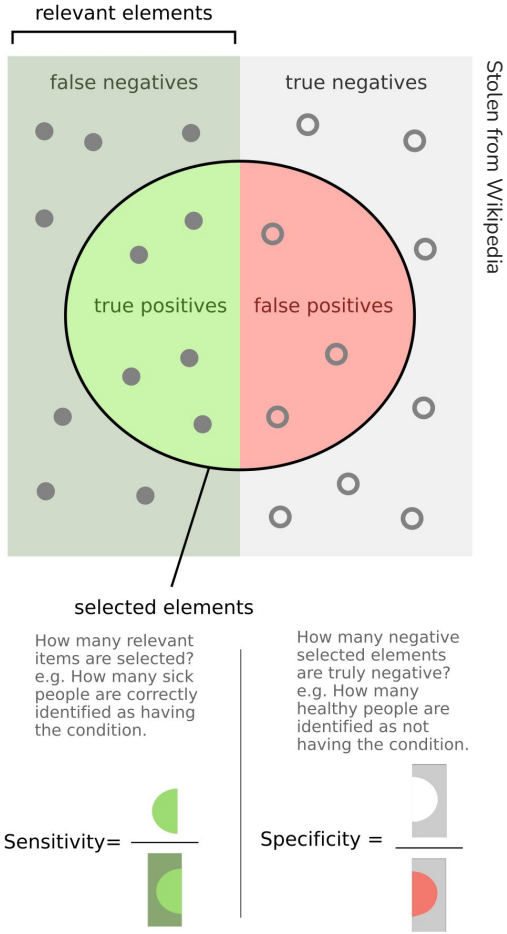
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Sensitivity & Specificity

What are the sensitivity and specificity of our model?

	Actually voted	Did not vote
Pred to vote	40	20
Pred stay home	15	25



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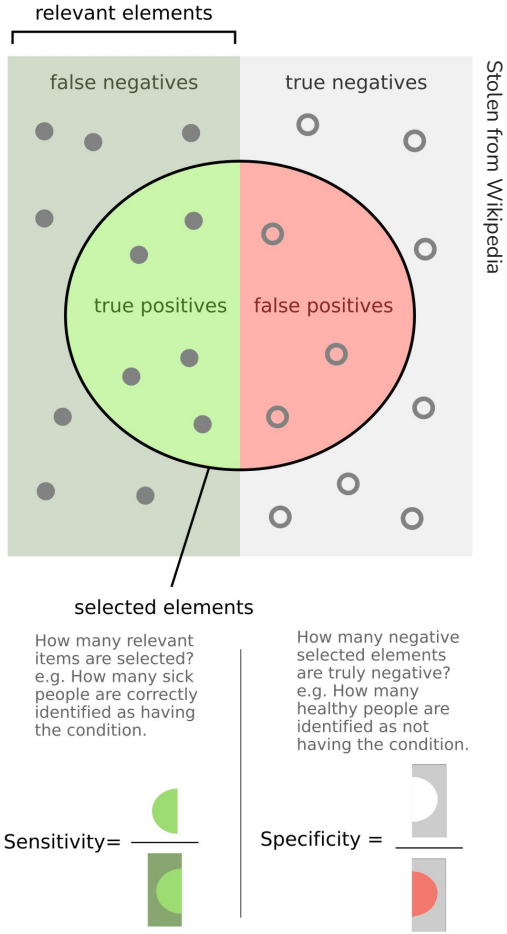
Sensitivity & Specificity

What are the sensitivity and specificity of our model?

	Actually voted	Did not vote
Pred to vote	40	20
Pred stay home	15	25

$$\text{Sensitivity} = \frac{40}{40+15} = \frac{40}{55} \approx 0.727$$

$$\text{Specificity} = \frac{25}{20+25} = \frac{25}{45} \approx 0.556$$

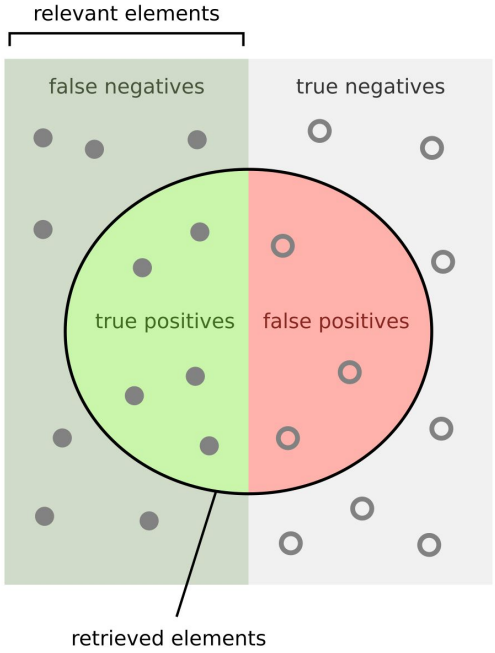


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Precision & Recall

The **precision**, aka **positive predictive value** - is, among all predicted positives, what proportion did were correct?

The **recall** is the **true positive rate** - that is... the same as **sensitivity**



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How many retrieved items are relevant?

Precision = $\frac{\text{true positives}}{\text{true positives} + \text{false positives}}$

How many relevant items are retrieved?

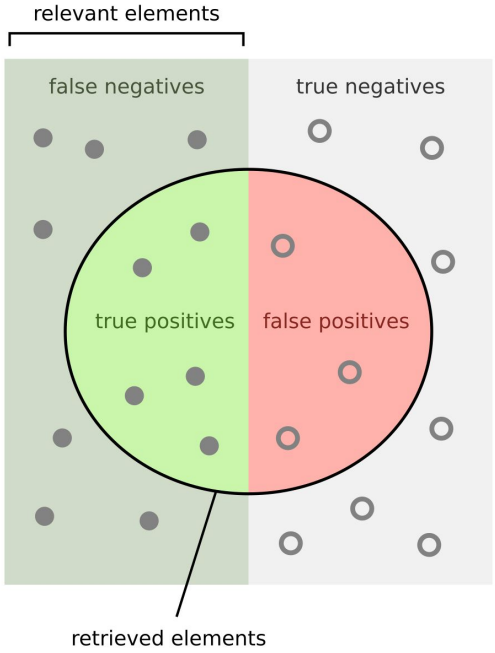
Recall = $\frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$

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Precision & Recall

What are the precision and recall of our model?

	Actually voted	Did not vote
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Stolen from Wikipedia

How many retrieved items are relevant?

Precision = 

How many relevant items are retrieved?

Recall = 

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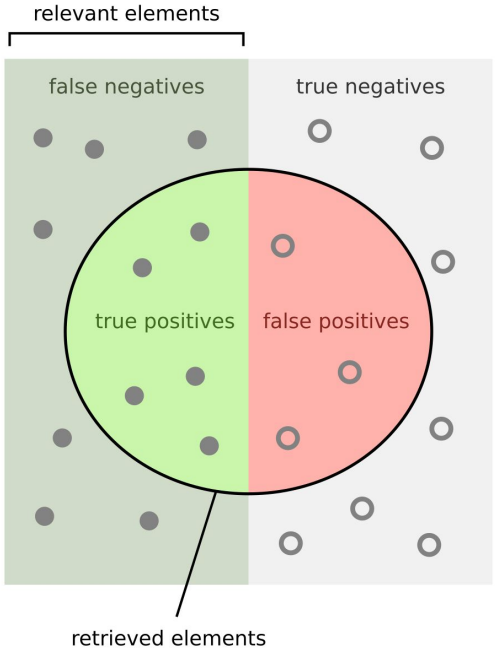
Precision & Recall

What are the precision and recall of our model?

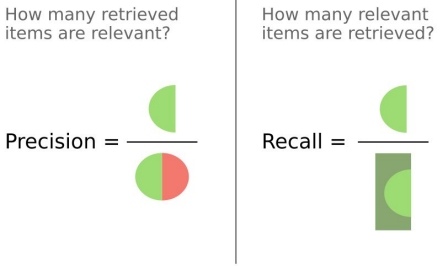
	Actually voted	Did not vote
Pred to vote	40	20
Pred stay home	15	25

$$\text{Precision} = \frac{40}{40+20} = \frac{40}{60} \approx 0.667$$

$$\text{Recall} = \frac{40}{40+15} = \frac{40}{55} \approx 0.727$$



Stolen from Wikipedia



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F_1 and F_β

Another single-number measure of model success is the **F_1 score**, which is an F-measure. You might use this when accuracy is a bad choice and you want to prioritize both precision and recall equally.

$$F_1 = \frac{2}{1/\text{recall} + 1/\text{precision}} = 2\frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

It is on a typical 0-to-1 scale, where 0 = bad and 1 = good

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F_1 and F_β

You could also compute an **F_β score**, which is the same as an F_1 score but gives a higher (or lower) priority to recall. For example, if $\beta = 2$, you believe that recall is twice as important as precision.

$$F_\beta = (1 + \beta^2)\frac{\text{precision} \times \text{recall}}{\beta^2 \text{precision} + \text{recall}}$$

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The ROC Curve and AUC

But First: A Case Study

Let’s return to the cancer screening problem. You’ve built a model to detect whether or not someone has cancer (positive) or not (negative). Consider the following predictions:

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But First: A Case Study

Let's return to the cancer screening problem. You've built a model to detect whether or not someone has cancer (positive) or not (negative). Consider the following predictions:

$$\hat{p} = 0.51$$

Cancer ✔

$$\hat{p} = 0.49$$

No Cancer ✗



Why is this wrong?

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Classification Thresholds

Depending on your problem, you might want to pick a **threshold** other than 0.5.

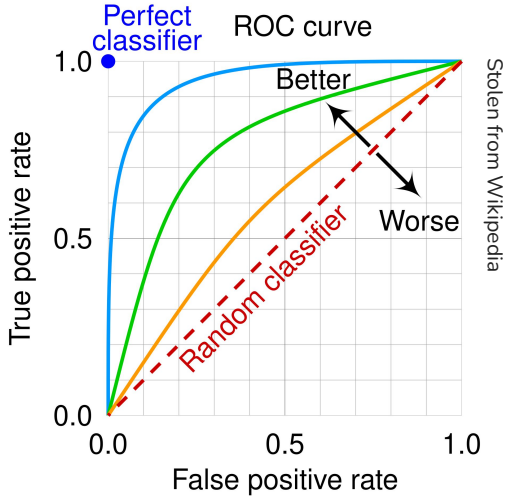
Modifying this can change (and maybe improve!) your confusion matrix.

We can plot the false positive rate and true positive rate on a graph *for each possible value of our threshold!*

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ROC Curve

The **Receiver Operating Characteristic (ROC) Curve** is the name of this graph. It's a line plot of all the possible TPR and FPR values for every possible value of the threshold.



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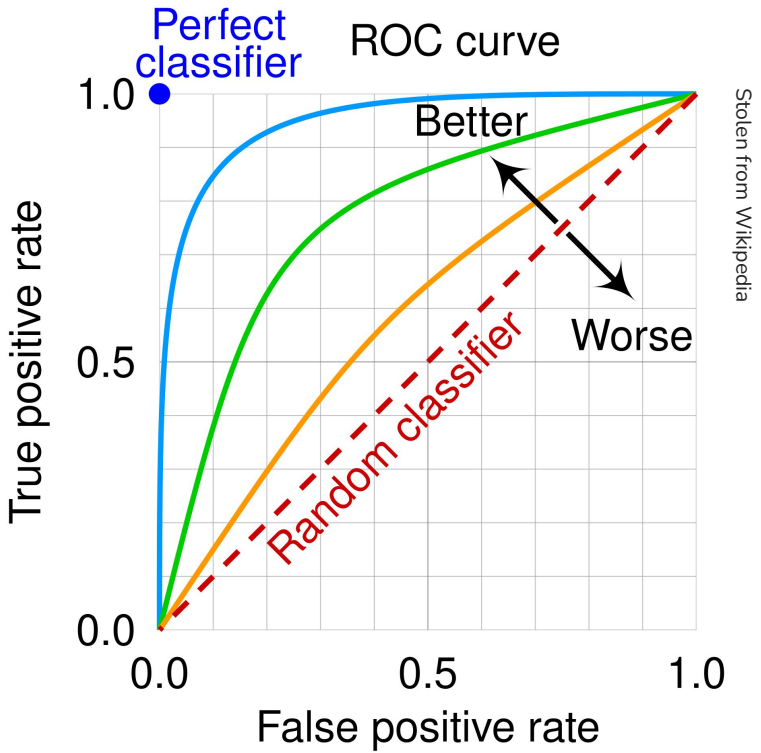
The AUC Score

The classifier that does total random guessing achieves an ROC curve at the 45 degree line.

Better classifiers “bow” outward towards the perfect classifier - one with 100% TPR and 0% FPR.

So, a metric we can derive from the ROC curve is to find the **area under the curve (AUC)**!

The AUC score has a maximum value of 1 and a minimum value of 0, but anything less than 0.5 would be below the “random guessing” line.



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What's a "positive"?

Notice that most of the classification metrics we've learned today reference something called a **positive** and a **negative**. But consider the following example:

Again, we're doing a cancer screening, but this time, there are three options:

1. No tumor
2. Benign tumor
3. Malignant tumor

What's the "positive" outcome?

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What's a "positive"?

There isn't one!



If we're doing multiclass classification, what metrics are useful to look at?

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Let's Go!



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